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H.W = 4

Pseudocode :

① Initialize parameters
- set weights $\omega$ to initial values.

② Defining logistic regression cost function (objective function)
- for each data point, compute the linear combination of weights and features.
$Z = \omega[0] + \omega[1] * propellant\_age + \omega[2] * storage\_temp$

③ Define gradient of cost function.
$$h_i = \frac{1}{1+e^{-Z}}$$
- compute the gradient of the cost function with respect to each weight.
- Return the gradient.

④ Define the line search function.
- ~~Beta~~ = set beta = 0.1 , tau = 0.5
- Return the appropriate step size.

⑤ Perform gradient descent
Set : maximum iterations = 1000000
epsilon = $10^{-3}$

- Repeat until convergence or maximum iterations.
- compute gradient of the cost function and check if gradient norm is below epsilon
· If yes , stop iteration
- Perform line search to find the optimal step size.
- update weights and print iteration number and gradient norm.

⑥ print minimum cost and optimal weights.
Printing the number of iteration and plotting the result (scatter plot and decision boundary contour).

⑦ save and display the plot.